MARYLEBONE ROSTER · LINKS DESIGNER

# Proposed new CEA link

How this rota would work from 1 January 2027, and how it compares with today's link

Prepared 7 Nov 2026 · every figure but the description under the name is calculated from the rota

PROPOSAL

## Balanced V9

**A 15-minute handover every weekday, from the three 07:30–16:00s to the three 15:45 closers, never more than three early or three late weeks in a row outside the cover weeks, no fatigue warning in any cover-week placement tested, and 14 days' leave buys 30 days off, as today**

| | |
|---|---|
| CODE | **MB-26-V9** |
| FINGERPRINT | **e2b7d945** |
| SET AGAINST | today's 20-week link |
| CREATED | 7 Nov 2026 |

**IN SHORT**

Balanced V9 meets all 9 December staffing rules and stays inside every hard limit. For staff, the biggest gains are about 14 full weekends off a year (today 10) and never more than 5 days in a row in the fixed duties (today 7). The main trade-offs: more single rest days, which are not a two-day break (about 12 a year each, today 10); and more late finishes (about 42 a year each, today 39).

**For discussion, not a decision.** This proposal has not yet been through the roster office. Still to settle: which link is used — colleagues' views come first — and then who starts on which line; the rota does not say who works which week.

**What it takes:** 26 people, 6 more than today's 20 · 5 cover weeks for leave and sickness · 10 on duty every Sunday (today 8; the rules ask for 10) — 83h 30m of Sunday overtime across the whole link each week (today 68h 15m).

## Can it work?

Does it meet the December staffing rules?

**9 of 9**

**rules met**

today's link meets 4 · each rule on page 6

Can it be run within the hard limits?

**Yes**

**inside every hard limit**

shortest gap 14h 20m (limit 12h) · most days in a row 5, up to 6 if a cover week falls badly (limit 13) · 35-hour contract exact · 218.6 days a year (limit 219)

Any avoidable fatigue warnings?

**0**

**none in the fixed duties**

today's link has 4 · today up to 5 if a cover week falls badly · guidance, not a pass or fail · page 7

Does staffing follow the trains better?

**Yes**

**closer than today, weekdays and weekend**

weekday · Sat · Sun 40.0 · 17.3 · 29.2 (today 52.0 · 65.9 · 70.2) · on weekdays a score about three-quarters of today's · page 5

## What it would mean for staff

### Likely positives for staff

- About 14 full weekends off a year (today 10), 7 in 26 weeks, never more than 5 weeks apart (today up to 7)
- Never more than 5 days in a row in the fixed duties (today 7)
- Four full weeks off in a row, Sunday to Saturday, for 13 days' leave (today 14)
- Shortest rest between shifts 14h 20m in the fixed duties (today 12h 30m)
- No avoidable fatigue warnings in the fixed duties (today 4)

*More detail on page 2*

### What colleagues may be concerned about

- More single rest days, which are not a two-day break — about 12 a year each (today 10)
- More late finishes — about 42 a year each (today 39)
- 8 new shift times to learn

AGAINST TODAY'S LINK

# What changes, in numbers

Balanced V9's figures are marked against today's 20-week link. ✓ **green**: better than today on something the rules, hard limits or fatigue guidance aim for, or on a plain gain such as full weekends off or the fit to the trains · ▲ **amber**: worse on one of those · ✕ **red**: a rule or limit broken · **unshaded**: a fact, or for colleagues to weigh.

| QUESTION | TODAY | BALANCED V9 | IN PLAIN WORDS |
|---|---|---|---|
| **STAFFING** | | | |
| People on the link | 20 | **26** | 6 more people, one per line |
| On duty each day<br>weekday · Sat · Sun | 11–12 · 10 · 8 | ✓ **15 · 14 · 10** | the December levels ask for 10 on a Sunday — met |
| On at the open<br>weekday · Sat · Sun | 4 · 4 · 3 | ✓ **4 · 5 · 4** | at least 4 every day — met |
| Still on duty at 22:00<br>weekday · Sat · Sun | 4 · 2 · 3 | ✓ **5 · 7 · 5** | at least 5 every day — met |
| Through to the close<br>weekday · Sat · Sun | 2 · 2 · 3 | ✓ **3 · 3 · 3** | at least 3 every day — met |
| How closely the floor follows the trains<br>weekday · Sat · Sun | 52.0 · 65.9 · 70.2 | ✓ **40.0 · 17.3 · 29.2** | lower is closer; weekdays about three-quarters of today's — page 5 |
| Cover weeks — for leave and sickness | lines 1, 7, 12, 17 | **lines 1, 6, 11, 16, 21** | gaps of 5, 5, 5, 5 and 6 weeks |
| Ticket office late shift on a Sunday — people | 1 | **2** | the December plan: both in the office 15:00–18:00, then one until 22:00; today one 14:30–23:25 closer keeps it until 22:00 |
| **WORKING PATTERN** | | | |
| Most days in a row, fixed duties | 7 | ✓ **5** | up to 6 (today 9) if a cover week falls badly; Chiltern's limit is 13 |
| Shortest rest between shifts | 12h 30m | ✓ **14h 20m** | in the fixed duties; the limit is 12 hours |
| Full weekends off, in the rotation | 4 in 20 | ✓ **7 in 26** | uneven: never more than 5 weeks apart (today up to 7) |
| Days at work a year, Mon–Sat | 219.0 | **218.6** | Sundays are overtime; a cover week is 4 days |
| Weeks on one shift time Mon–Fri, all earlies or all lates | 7 of 16 | **18 of 21** | and its Saturday or Sunday duty the same kind |
| Weeks mixing earlies and lates | 7 of 16 | **2 of 21** | a week with both early and late shifts in it |
| Longest block of earlies or lates | 2 (up to 3) | **3 (up to 4)** | weeks in a row on one kind; "up to" with the cover weeks |
| Single rest days, in the rotation | 4 in 20 | **6 in 26** | a rest day on its own: about 12 a year (today 10) |
| Six-day weeks | 1 | **0** | weeks with six days at work, Sundays counted |
| 14 days' leave buys at best · leave for four weeks off | 30 · 14 days | ✓ **30 · 13 days** | the most days off in a row for 14 days' leave · the least for four full weeks, Sun–Sat |
| **SHIFTS** | | | |
| Average shift · longest shift | 8h 14m · 9h 10m | **8h 16m · 9h 00m** | the average is Monday to Saturday; the longest is on any day |
| Early shifts, shortest to longest | 7h 15m–8h 30m | **7h 15m–9h 00m** | an early starts before 11:00 · any day of the week |
| Late shifts, shortest to longest | 7h 30m–9h 10m | **8h 00m–8h 55m** | a late starts at 11:00 or after · any day of the week |
| Closing shift<br>weekday · Sat · Sun | 8h 40m · 9h 10m · 8h 55m | ✓ **8h 10m · 8h 40m–8h 55m · 8h 10m** | the shift that locks up |
| How far the start time moves from one week to the next — on average | 4h 00m | ✓ **1h 51m** | smaller is easier on the body clock |
| Different shift times | 18 | ✓ **17 (9 worked today)** | 8 new to learn, on 15 of the 21 working weeks — listed on page 4 |
| **EACH PERSON'S YEAR, ON AVERAGE** | | | |
| Full weekends off | 10 | ✓ **14** | 27% of weeks (today 20%) |
| Finishing at 23:00 or later | 39 | **42** | about one extra every 17 or 18 weeks |
| Starting at 06:20, the open | 62 | **50** | about one fewer every 4 weeks |
| Shifts over 8h 30m, Mon–Sat | 31 | ✓ **8** | Sunday duties are overtime, so not counted |
| Saturdays worked | 26 | **28** | paid at time and a quarter; cover weeks left out |
| Sunday overtime to share | 21 | **20** | Sundays are overtime, as today |
| **FATIGUE AND THE RULES** | | | |
| Avoidable fatigue warnings, fixed duties | 4 | ✓ **0** | today up to 5 if a cover week falls badly; guidance only, page 7 |
| December staffing rules met | 4 of 9 | ✓ **9 of 9** | confirmed verbally, 29 Sep 2026 — page 6 |

**The workings:** the rota (page 3) · shift times (4) · staffing through the day (5) · the rules (6) · fatigue checks (7) · can I trust these numbers? (8)

THE ROTA

# The 26-line link

Sunday to Saturday per line; everyone moves down one line each week and line 26 goes back to line 1. Hours and days at the right; the number on duty each day beneath.

**21** working weeks · **5** cover weeks · **35h 00m** a week on average
at most **5** days in a row (up to 6 with a cover week) · **18** weeks on one shift time, all earlies or all lates

| LINE | SUN | MON | TUE | WED | THU | FRI | SAT | MON–SAT hours | ALL WEEK hours | DAYS WORKED Mon–Sat | TURNS Mon–Fri |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | SP | SP | SP | SP | SP | SP | SP | *35h 00m* | *35h 00m* | — | — |
| 2 | 15:15<br>23:25 | 15:45<br>23:55 | RD | RD | 15:45<br>23:55 | 15:45<br>23:55 | 15:00<br>23:55 | 33h 25m | 41h 35m | 4 | **1** |
| 3 | 15:15<br>23:25 | RD | 15:45<br>23:55 | 15:45<br>23:55 | 15:45<br>23:55 | 15:45<br>23:55 | RD | 32h 40m | 40h 50m | 4 | **1** |
| 4 | RD | 15:45<br>23:55 | 15:45<br>23:55 | 15:45<br>23:55 | RD | 15:45<br>23:55 | 15:00<br>23:55 | 41h 35m | 41h 35m | 5 | **1** |
| 5 | 15:15<br>23:25 | 14:00<br>22:30 | 14:00<br>22:30 | RD | RD | RD | 08:00<br>16:30 | 25h 30m | 33h 40m | 3 | 1 |
| 6 | SP | SP | SP | SP | SP | SP | SP | *35h 00m* | *35h 00m* | — | — |
| 7 | 07:15<br>15:45 | 07:30<br>16:00 | RD | RD | RD | 07:30<br>16:00 | 06:20<br>15:00 | 25h 40m | 34h 10m | 3 | **1** |
| 8 | 07:15<br>15:45 | 06:20<br>13:45 | 06:20<br>13:45 | RD | RD | 06:20<br>13:45 | RD | 22h 15m | 30h 45m | 3 | **1** |
| 9 | RD | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | 12:00<br>20:30 | RD | 42h 30m | 42h 30m | 5 | **1** |
| 10 | RD | 13:30<br>22:00 | 13:30<br>22:00 | 13:30<br>22:00 | RD | RD | 14:00<br>22:30 | 34h 00m | 34h 00m | 4 | **1** |
| 11 | SP | SP | SP | SP | SP | SP | SP | *35h 00m* | *35h 00m* | — | — |
| 12 | 09:00<br>18:00 | RD | RD | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:50 | 32h 30m | 41h 30m | 4 | **1** |
| 13 | RD | RD | 13:30<br>22:00 | 13:30<br>22:00 | 13:30<br>22:00 | 13:30<br>22:00 | 14:30<br>22:30 | 42h 00m | 42h 00m | 5 | **1** |
| 14 | RD | RD | 14:00<br>22:30 | 14:00<br>22:30 | 14:00<br>22:30 | 14:00<br>22:30 | RD | 34h 00m | 34h 00m | 4 | **1** |
| 15 | RD | 15:45<br>23:55 | 15:45<br>23:55 | 15:45<br>23:55 | 15:45<br>23:55 | RD | 15:15<br>23:55 | 41h 20m | 41h 20m | 5 | **1** |
| 16 | SP | SP | SP | SP | SP | SP | SP | *35h 00m* | *35h 00m* | — | — |
| 17 | 07:15<br>15:45 | 06:20<br>14:20 | RD | RD | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:50 | 32h 30m | 41h 00m | 4 | **1** |
| 18 | RD | RD | 06:20<br>14:20 | 07:30<br>16:00 | 06:20<br>13:45 | 07:30<br>16:00 | 06:20<br>14:50 | 40h 55m | 40h 55m | 5 | 3 |
| 19 | RD | RD | 06:20<br>13:35 | 06:20<br>13:35 | 06:20<br>13:35 | 06:20<br>13:35 | RD | 29h 00m | 29h 00m | 4 | **1** |
| 20 | RD | 06:20<br>13:35 | 07:30<br>16:00 | 06:20<br>13:45 | 07:30<br>16:00 | RD | 14:00<br>22:30 | 40h 10m | 40h 10m | 5 | 3 |
| 21 | SP | SP | SP | SP | SP | SP | SP | *35h 00m* | *35h 00m* | — | — |
| 22 | 14:30<br>22:30 | 13:30<br>22:00 | RD | RD | 13:30<br>22:00 | 13:30<br>22:00 | 14:30<br>22:30 | 33h 30m | 41h 30m | 4 | **1** |
| 23 | 14:30<br>22:30 | 14:00<br>22:30 | RD | 14:00<br>22:30 | 14:00<br>22:30 | 14:00<br>22:30 | RD | 34h 00m | 42h 00m | 4 | **1** |
| 24 | RD | 07:30<br>16:00 | 07:30<br>16:00 | 07:30<br>16:00 | 07:30<br>16:00 | 07:30<br>16:00 | RD | 42h 30m | 42h 30m | 5 | **1** |
| 25 | RD | 07:30<br>16:00 | 07:30<br>16:00 | 07:30<br>16:00 | 07:30<br>16:00 | RD | 06:20<br>14:50 | 42h 30m | 42h 30m | 5 | **1** |
| 26 | 07:15<br>15:45 | 06:20<br>14:20 | 06:20<br>14:20 | 06:20<br>14:20 | RD | RD | 08:00<br>16:30 | 32h 30m | 41h 00m | 4 | **1** |
| **On duty** | **10**<br>SP 5 | **15**<br>SP 5 | **15**<br>SP 5 | **15**<br>SP 5 | **15**<br>SP 5 | **15**<br>SP 5 | **14**<br>SP 5 | 35h 00m | 38h 13m | 4.24 | 18<br>one-turn weeks |
| **Minutes** | 5,010 | 7,390 | 7,390 | 7,390 | 7,390 | 7,390 | 7,150 | **44,100** | **49,110** | | |

Early turns · Late turns · Cover (spare) week — four duties Monday to Saturday, any turn; its Sunday can be overtime · Rest day

Mon–Sat hours average 35h 00m over 26 lines, a cover week counted as a contracted week · days worked average over the 21 working lines · All week: grey = no Sunday duty · Turns: how many different shift times the week works Monday to Friday; **bold** marks a one-turn week (see below) · Minutes: total duty minutes each day — Monday to Saturday adds up to the contract, 44,100 minutes: 35 hours for each of the 21 working weeks

## The words on this page

**Link** — the whole rota: 26 weeks laid out as 26 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 26 people.

**Line** — one week of the pattern, Sunday to Saturday. Everyone moves down a line each week, and line 26 goes back to line 1.

**Turn** — a shift time, such as 06:20–13:35. An **early** starts before 11:00; a **late** starts at 11:00 or later.

**Rest day (RD)** — a day off inside the contracted week. Two together give a longer break, always more than 48 hours. One on its own usually gives less, though the hours off depend on the shifts either side.

**Cover (spare) week** — a week with no fixed times. Whoever is on it works four duties, Monday to Saturday, placed by the roster clerk to cover leave and sickness; its Sunday can be overtime on top.

**Sunday** — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon–Sat and all week.

**The contract** — 35 hours a week, averaged over the 26 lines; a cover week counts as a full contracted week.

**One-turn week** — one clock time Monday to Friday, with every day of the week an early or every day a late (shown bold in the Turns column); the easiest kind of week to live around.

**The floor** — staff out on the station (gates, concourse, platforms); the ticket office joins it only at the quiet ends (page 5).

**Opener / closer** — the first people on duty in the morning and the last at night.

SHIFT TIMES AND THE SHAPE OF A WEEK

# Shift times

The shift times staff would need to learn, every shift time beside the ones worked today, and the shape of a working week.

**17**
shift times in all

**9**
already worked today

**8**
are new — nobody works them today

## The new times to learn

**06:20–15:00**
8h 40m · early · Saturdays

**07:30–16:00**
8h 30m · early · weekdays

**09:00–18:00**
9h 00m · early · Sundays

**12:00–20:30**
8h 30m · late · weekdays

**14:30–22:30**
8h 00m · late · Saturdays and Sundays

**15:00–23:55**
8h 55m · late · Saturdays

**15:15–23:25**
8h 10m · late · Sundays

**15:45–23:55**
8h 10m · late · weekdays

**No longer used:** 9 of today's 18 shift times — 06:20–14:00, 08:30–16:30, 11:00–19:30, 12:00–20:00, 13:00–21:00, 13:30–21:00, 14:30–22:00, 14:30–23:25, 14:45–23:55.

## Every shift time, beside today's

| | | | TODAY'S LINK | | | PROPOSED | | |
|---|---|---|---|---|---|---|---|---|
| SHIFT TIME | | | MON–FRI | SAT | SUN | MON–FRI | SAT | SUN |
| 06:20–13:35 | | 7h 15m · Early | 1 | | | **1** | | |
| 06:20–13:45 | | 7h 25m · Early | 1 | | | **1** | | |
| *06:20–14:00* | DROPPED | *7h 40m · Early* | | *1* | | | ***0*** *(−1)* | |
| 06:20–14:20 | | 8h 00m · Early | 2 | | | **2** | | |
| 06:20–14:50 | | 8h 30m · Early | | 3 | | | **4** (+1) | |
| **06:20–15:00** | NEW | 8h 40m · Early | | | | | **1** (+1) | |
| 07:15–15:45 | | 8h 30m · Early | | | 3 | | | **4** (+1) |
| **07:30–16:00** | NEW | 8h 30m · Early | | | | **3** (+3) | | |
| 08:00–16:30 | | 8h 30m · Early | 2 | | | **0** (−2) | **2** (+2) | |
| *08:30–16:30* | DROPPED | *8h 00m · Early* | | | *1* | | | ***0*** *(−1)* |
| **09:00–18:00** | NEW | 9h 00m · Early | | | | | | **1** (+1) |
| *11:00–19:30* | DROPPED | *8h 30m · Late* | *1* | | | ***0*** *(−1)* | | |
| *12:00–20:00* | DROPPED | *8h 00m · Late* | | *1* | | | ***0*** *(−1)* | |
| **12:00–20:30** | NEW | 8h 30m · Late | | | | **1** (+1) | | |
| *13:00–21:00* | DROPPED | *8h 00m · Late* | | | *1* | | | ***0*** *(−1)* |
| *13:30–21:00* | DROPPED | *7h 30m · Late* | | *1* | | | ***0*** *(−1)* | |
| 13:30–22:00 | | 8h 30m · Late | 1 | | | **2** (+1) | | |
| 14:00–22:30 | | 8h 30m · Late | 2 | | | **2** | **2** (+2) | |
| *14:30–22:00* | DROPPED | *7h 30m · Late* | | *2* | | | ***0*** *(−2)* | |
| **14:30–22:30** | NEW | 8h 00m · Late | | | | | **2** (+2) | **2** (+2) |
| *14:30–23:25* | DROPPED | *8h 55m · Late* | | | *3* | | | ***0*** *(−3)* |
| *14:45–23:55* | DROPPED | *9h 10m · Late* | | *2* | | | ***0*** *(−2)* | |
| **15:00–23:55** | NEW | 8h 55m · Late | | | | | **2** (+2) | |
| **15:15–23:25** | NEW | 8h 10m · Late | | | | | | **3** (+3) |
| 15:15–23:55 | | 8h 40m · Late | 2 | | | **0** (−2) | **1** (+1) | |
| **15:45–23:55** | NEW | 8h 10m · Late | | | | **3** (+3) | | |

People on each shift time on a Saturday, a Sunday and — under Mon–Fri — the weekday with most of that time, so Mon–Fri can add up to more than work on one day. **(+2)** and **(−1)**: more or fewer people on that time than today — a difference, not a verdict. **New**: nobody works it today. **Dropped**: worked today, not in this link.

Monday to Friday add up to 36,950 minutes of duty and Saturday 7,150 — together exactly 44,100 minutes — 35 hours for each of the 21 working weeks, the contract (a cover week counts as a full 35-hour week on top). Sunday adds 5,010, paid on top of the contracted week. 8 of the 17 shift times are new — nobody works them today.

### The shape of a week

| MEASURE | TODAY | PROPOSED |
|---|---|---|
| Rest breaks of two days or more<br>of all the breaks between duties | 13 of 17 | **20 of 26** |
| Single rest days<br>a rest day with a worked day either side | 4<br>(1 beside a cover week) | **6** |
| Working weeks by days worked, Sunday in<br>cover weeks left out | 1 × 3-day<br>3 × 4-day<br>11 × 5-day<br>1 × 6-day | **6 × 4-day**<br>**15 × 5-day** |
| Working weeks with a Sunday<br>the number on duty each Sunday — all overtime | 8 of 16 | **10 of 21** |

STAFFING THROUGH THE DAY

# People on duty, hour by hour

Today's link and the proposal, hour by hour, against the December 2026 timetable. A darker orange hour carries more of the day's trains.

| How closely the floor follows the trains | Weekdays | Saturday | Sunday |
|---|---|---|---|
| Today's link | 52.0 | 65.9 | 70.2 |
| **Balanced V9** | ✓ 40.0 | ✓ 17.3 | ✓ 29.2 |

**The floor follows the trains more closely than today** — scores about three-quarters of today's on weekdays, a quarter on Saturday and two-fifths on Sunday. Lower is closer; 0 is a perfect match.

**People on duty** in that hour — darker blue is more. Anyone on for any part of the hour counts, so a handover counts both people.

**Read down each column:** the most people should be under the darkest orange.

**Trains** — carriages arriving and leaving in that hour; darker is busier. A guide from the timetable, not passenger numbers.

**Strongest rows:** the floor and the trains — the two the match compares.

## Monday to Friday staffed 06:20–23:55 · a row for each different weekday

| | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | Match |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Today, Mon · Wed — all on duty** | | 4 | 4 | 6 | 6 | 6 | 7 | 7 | 7 | 7 | 7 | 7 | 5 | 5 | 5 | 4 | 4 | 4 | 2 | |
| **Today, Tue · Thu · Fri — all on duty** | | 4 | 4 | 6 | 6 | 6 | 7 | 7 | 8 | 8 | 8 | 8 | 6 | 6 | 6 | 5 | 5 | 4 | 2 | |
| of whom in the ticket office | | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | | |
| **of whom on the floor Mon · Wed** | | 3 | 3 | 4 | 4 | 4 | 5 | 5 | 5 | 3 | 5 | 5 | 3 | 3 | 4 | 3 | 3 | 3 | 2 | 61.1 |
| **of whom on the floor Tue · Thu · Fri** | | 3 | 3 | 4 | 4 | 4 | 5 | 5 | 6 | 4 | 6 | 6 | 4 | 4 | 5 | 4 | 4 | 3 | 2 | 50.1 |
| **Proposed — all on duty** | | 4 | 7 | 7 | 7 | 7 | 7 | 8 | 10 | 10 | 11 | 8 | 8 | 8 | 8 | 8 | 7 | 5 | 3 | |
| of whom in the ticket office | | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | | |
| **of whom on the floor** | | 3 | 6 | 5 | 5 | 5 | 5 | 6 | 8 | 6 | 9 | 6 | 6 | 6 | 7 | 7 | 6 | 4 | 3 | 40.0 |
| **Trains — carriages in and out** | 5 | 69 | 107 | 127 | 122 | 95 | 85 | 75 | 84 | 79 | 87 | 105 | 140 | 138 | 107 | 90 | 73 | 75 | 66 | |

## Saturday staffed 06:20–23:55

| | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | Match |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Today — all on duty** | | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 6 | 9 | 6 | 6 | 6 | 6 | 6 | 5 | 4 | 2 | 2 | |
| of whom in the ticket office | | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | | | |
| **of whom on the floor** | | 3 | 3 | 2 | 2 | 2 | 2 | 3 | 4 | 5 | 4 | 4 | 4 | 4 | 5 | 4 | 3 | 2 | 2 | 65.9 |
| **Proposed — all on duty** | | 5 | 5 | 7 | 7 | 7 | 7 | 7 | 7 | 11 | 9 | 9 | 7 | 7 | 7 | 7 | 7 | 7 | 3 | |
| of whom in the ticket office | | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | | |
| **of whom on the floor** | | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 5 | 7 | 7 | 7 | 5 | 5 | 6 | 6 | 6 | 6 | 3 | 17.3 |
| **Trains — carriages in and out** | 6 | 24 | 58 | 75 | 81 | 92 | 88 | 77 | 64 | 68 | 72 | 84 | 82 | 67 | 64 | 79 | 78 | 62 | 41 | |

## Sunday staffed 07:15–23:25

| | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | Match |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Today — all on duty** | | | 3 | 4 | 4 | 4 | 4 | 4 | 5 | 8 | 8 | 5 | 4 | 4 | 4 | 4 | 3 | 3 | 3 | |
| of whom in the ticket office | | | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 1 | 1 | 1 | 1 | 1 | 1 | | | |
| **of whom on the floor** | | | 2 | 3 | 2 | 2 | 2 | 2 | 3 | 5 | 5 | 4 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 70.2 |
| **Proposed — all on duty** | | | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 7 | 10 | 6 | 6 | 5 | 5 | 5 | 5 | 5 | 3 | |
| of whom in the ticket office | | | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 4 | 2 | 2 | 1 | 1 | 1 | 1 | | | |
| **of whom on the floor** | | | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 5 | 6 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 3 | 29.2 |
| **Trains — carriages in and out** | | | 15 | 45 | 65 | 71 | 74 | 71 | 61 | 55 | 64 | 69 | 78 | 73 | 75 | 52 | 71 | 65 | 50 | |

## Good to know

- **The ticket office.** In the proposal, two people on every ticket-office shift, and not floor cover except at the quiet ends: Monday to Saturday one of each pair is on the floor until 08:00 and from 19:30, so the office row drops to 1 in those hours and the floor row gains one; where it shows 4, both pairs are there at the changeover. On a Saturday its lates stay until 22:30, to cash up after it shuts at 22:10; today's roster has them leaving at 22:00. The "of whom" rows take the office out of each side, so the floor rows compare like with like.
- **Sunday.** One early is on the floor until 09:00; the two lates are on the floor until 15:00, then both in the office — at least 30 minutes' handover — and one goes back to the floor from 18:00, the other at 22:00, once the office has shut at 21:40 and cashed up; a late that runs to 22:30 spends its last half hour on the floor. Today one of the three 14:30–23:25 closers keeps the office until 22:00, so in the 16:00 and 17:00 hours today's office row reads 1 where the proposal's reads 2 (the 15:00 hour also counts the earlies finishing), and from 22:00 neither has anyone in the office.
- **Reading the match.** The weekday figure at the top is worked out from the five weekdays together, not averaged from the day scores. It measures **shape, not numbers** — each hour's share of the day's staff against its share of the day's trains — so an extra person in a quiet hour makes it worse even though nobody is worse off. Read it beside the numbers of people, never alone.
- **Cover weeks.** They are not in these figures, because their times are not known yet; actual staffing depends on where those duties are used.
- **Sunday's last trains.** On Sundays five trains in the December 2026 timetable arrive or leave after the 23:25 finish (the last at 23:54); the bar under the Sunday 23:00 hour marks the time after the finish. No duty runs past 23:25 on a Sunday — that is agreed practice and stays so (settled 28 Sep 2026) — so those trains fall outside the staffed day of any link, today's included.

THE RULES

# Hard limits and the December 2026 rules

Three hard limits a rota must meet to be run at all, and the ceiling of 219 contracted days a year, then the nine December 2026 staffing rules, each with today's link beside it.

## Hard limits — met, or the rota cannot be run

### MOST DAYS IN A ROW

**6**

✓ **met · longest possible · limit 13 · today 9 · fixed duties 5 (today 7)**

Longest possible run is 6 days, within Chiltern's limit of 13. A cover week is 4 duties on Monday to Saturday, placed by the roster clerk, and its Sunday can be worked as overtime on top, so this is the worst case the link allows. Worked as a block of four rather than split day-on-day-off, the worst case is the same — **6** days. It does depend on where the roster clerk puts the four, and on the Sunday: placed well, with no Sunday overtime, the longest run is **5** days.
The written source of the 13-day limit is still to be confirmed.

### SHORTEST REST BETWEEN SHIFTS

**14h 20m**

✓ **met · in the fixed duties · limit 12h · today 12h 30m**

Tightest anywhere: **14h 20m — line 7 Fri 07:30–16:00 into Sat 06:20–15:00** (today's tightest: 12h 30m).

### THE CONTRACT

**35h 00m**

✓ **met · a week, Monday to Saturday, on average · must be exact · 218.6 contracted days a year, 219 at most**

735h of duty a week across 21 working lines. Sunday duties (83h 30m in all) are paid on top as rest-day working, as they are today. Individual weeks range 22h 15m–42h 30m; only the average is the contract.

## The December 2026 rules

✓ 9 of 9 met

| RULE | TODAY'S LINK | THIS PROPOSAL | WHAT THE RULE ASKS |
|---|---|---|---|
| At least four on at the open, every day | ✗ 4 · 4 · 3 | ✓ 4 · 5 · 4 | from 06:20 (07:15 on a Sunday) |
| At least three through to the close, every day | ✗ 2 · 2 · 3 | ✓ 3 · 3 · 3 | until 23:55 (23:25 on a Sunday) |
| At least five still on duty at 22:00, every day | ✗ 4 · 2 · 3 | ✓ 5 · 7 · 5 | someone finishing at 22:00 does not count |
| Ten on a Sunday | ✗ 8 | ✓ 10 | every duty rostered that day counts |
| The ticket office: two early and two late, identical shifts, every day | ✗ Mon–Fri yes · Sat no · Sun no | ✓ Mon–Fri yes · Sat yes · Sun yes | Mon–Fri 06:20–14:20 and 14:00–22:30; Sat 06:20–14:50 and 14:30–22:30; Sun two from 07:15 and two to 22:00 or 22:30 |
| At least two on the floor at every moment | ✓ fewest 2 · 2 · 2 — first at Mon 22:30, Sat 08:00, Sun 07:15 | ✓ fewest 3 · 3 · 3 — first at Mon 06:20, Sat 22:30, Sun 07:15 | the ticket office counts only while its staff help on the floor — Mon–Sat one of each pair to 08:00 and from 19:30; Sunday one early to 09:00, both lates to 15:00, one from 18:00, both from 22:00; checked every five minutes |
| Handovers: 15 minutes to each closer, 20 in the ticket office (30 on a Sunday) | ✓ floor met · ticket office 20 min | ✓ floor met · ticket office 20 min | the Sunday office handover runs from 15:00, when the lates join it; on a Sunday each opener also stays until 15 minutes after the last closer arrives; where the design does not roster the office pairs, the plan's posts are assumed |
| Sunday duties between 8h and 9h | ✓ 8h 00m–8h 55m | ✓ 8h 00m–9h 00m | every Sunday duty, start to finish |
| No more shift times than today | ✓ 18 | ✓ 17 | today 18 |

FATIGUE CHECKS

# Fatigue checks

From the Office of Rail and Road's good-practice guidance, *Fatigue Factors*, page 3 (December 2021): 21 roster patterns that tend to tire people, with 4 checks from the rail industry's fatigue guidance (MRSF). ⚠ **present** — worth a look, not a breach · ✓ **clear** · ● **standing** — comes with the station's hours or any weekly link.

**AVOIDABLE FATIGUE WARNINGS**

**0** in the fixed duties · today's link has 4 · today up to 5 if a cover week falls badly · each is a question to discuss, not a breach

**STANDING — COME WITH EVERY WEEKLY LINK**

**2** early starts at a 06:20 station, and a rotation that moves everyone one line a week · recorded, not counted

For each of the 25 patterns this table asks whether it is in the rotation, today and proposed, and how big it is. A design showing nothing is not thereby approved: this is guidance, not a fatigue risk assessment.

| PATTERN | TODAY'S LINK | PROPOSED |
|---|---|---|
| **Early starts, 05:00 to 06:59**<br>FF2 · Early shift starting 05:00 to 06:59 · Time of day | ● 24 | ● 25 |
| **Very early starts, before 05:00**<br>FF3 · Very early shift starting before 05:00 · Time of day | ✓ 0 | ✓ 0 |
| **Shifts over 12 hours**<br>FF5 · Day shift over 12h · Duty length | ✓ 0 | ✓ 0 |
| **Early shifts over 10 hours**<br>FF7 · Early shift over 10h · Duty length | ✓ 0 | ✓ 0 |
| **Too little rest after a run of early starts**<br>FF8b · Fewer than 2 days rest after a block of early starts · Recovery time | ⚠ 2<br>⚠ up to 3 if a cover week falls badly | ✓ 0 |
| **Too many shifts without a two-day break**<br>FF11 · More than 13 consecutive shifts without a 48h break · Recovery time | ✓ 8<br>✓ up to 13 if a cover week falls badly | ✓ 9<br>✓ up to 12 if a cover week falls badly |
| **Runs of 12-hour day shifts**<br>FF10 · More than 4 consecutive 12h day shifts · Cumulative | ✓ 0 | ✓ 0 |
| **Long runs of early shifts**<br>FF15 · More than 4 consecutive early shifts in a rotating pattern · Cumulative | ⚠ 5<br>⚠ up to 7 if a cover week falls badly | ✓ 4 |
| **Long runs of day shifts**<br>MRSF · More than 12 consecutive day shifts · Cumulative | ✓ 7<br>✓ up to 9 if a cover week falls badly | ✓ 5<br>✓ up to 6 if a cover week falls badly |
| **More than 55 hours in any seven-day period**<br>MRSF · More than 55 hours worked in any 7-day period · Cumulative | ⚠ 60h 45m | ✓ 51h 00m |
| **Long runs of 8-hour shifts**<br>MRSF · More than 7 consecutive 8h shifts · Cumulative | ✓ 7<br>⚠ up to 9 if a cover week falls badly | ✓ 5<br>✓ up to 6 if a cover week falls badly |
| **Start times moving earlier through the week**<br>FF17 · Backward rotating pattern · Circadian | ✓ 10 backward / 12 forward | ✓ 12 backward / 12 forward |
| **Changing shift type about once a week**<br>FF18 · Rotating pattern of about a week · Circadian | ● 20-line rotation ·<br>typically 4h 00m a week | ● 26-line rotation ·<br>typically 1h 51m a week |
| **Start times that jump by more than two hours**<br>FF19 · Successive start times varying by more than 2 hours · (definition to confirm) · Circadian | ⚠ 6 | ✓ 0 |
| **Less than 12 hours between two shifts**<br>FF13 · Less than 12h rest in any 24h · Intervals between duties | ✓ 0 | ✓ 0 |

Also on the list, and applying to neither link: time of day (FF1) · duty length (FF4) · cumulative (MRSF) · night shifts (FF6, FF8, FF9, FF12, FF14, FF16, FF20).

- **Standing factors (●) are recorded for information, not counted.** FF2 comes with the station's 06:20 open, and FF18 — the ORR's concern with a rotation that changes shift type about once a week (Managing rail staff fatigue, 7.68) — comes with every weekly link, which moves everyone one line a week by construction; its week-to-week step is shown as information.
- **How to read the table.** Each mark and figure is for the duties the rota fixes, the cover weeks left out because their duties are not known yet, for today's link as for the proposal. Where a cover week's four duties could make a row worse, "up to" beneath it gives the worst place they could fall. FF2, FF7, FF8b and FF15 use the ORR's early, a start from 05:00 to 06:59, not this sheet's 11:00. FF11 counts shifts between two-day breaks: two rest days together, always over 48 hours off. A single rest day is not counted even when it gives 48 hours, so FF11 can only overstate a run, and can exceed the most days worked in a row.
- **Cover weeks.** A cover week's four duties can fall on Monday to Saturday in different ways, and its Sunday can be worked as overtime on top; on this design that moves the most days worked in a row (5 if placed well, with no Sunday overtime), the run of 8-hour shifts (5 if placed well, with no Sunday overtime), FF11 (12 at worst, 9 if the four are worked together with no Sunday overtime) and the 55-hour row only if its duties are longer than 8 hours: 52h 00m with the four contracted duties at 8h 45m, the contract's 35 hours, and the Sunday at 8 hours, still under 55. Where the four fall, and whether the Sunday is worked, are for the roster office. This page is an aid to discussion, not a fatigue risk assessment.

THE METHOD

# Can I trust these numbers?

How firm each figure is, what it is based on, and what no rota can tell you.

### Firm

Worked out exactly from the rota, and checked again by the app:

- the rest gaps and the most days in a row
- the 35-hour contract
- full weekends off
- how many people are on duty, hour by hour
- the shift times and their lengths

### A guide, not a verdict

Useful for comparing, never a pass or a fail:

- how closely the floor follows the trains
- the fatigue warnings — guidance from the rail regulator
- how familiar the shift times are
- figures "each, a year" — the fixed duties only, the cover weeks left out

### Not known yet

No rota can say these; they are for the people who work it and the managers who roster it:

- where each cover week's four duties fall
- how leave and sickness land
- whether staff accept the new shift times
- Wembley event days — the rota follows the trains, not the crowds
- who starts on which line
- the written source of the 13-day limit

## What each figure is based on

### Where the figures come from

Every figure is worked out from the rota on page 3, by the Marylebone Roster app's own checks where it has them and by the tooling beside them for the rest, and the same calculation is run on today's link. Only the one-line description under the name is written by hand.

### Cover weeks

Each is marked on all seven days but worked on four, Monday to Saturday, placed by the roster clerk; its Sunday can be worked as overtime on top. A cover week counts as a full contracted week and as four days at work. Its shift times are not known yet, so the runs, rest gaps, yearly counts and fatigue warnings are worked out on the fixed duties — the cover weeks left out, and today's link the same way. Where a cover week's four duties could make a figure worse, the worst place they could fall, with its Sunday worked, is given beside it as "up to", and the hard limits are tested at that worst case. Whatever is later given in a cover week must still be rostered within the normal limits.

### Sundays

Sunday is not in the contract: Sunday duties are overtime, as today. So "days at work" and the 35-hour week are Monday to Saturday. A Sunday off costs no annual leave.

### Each person, a year

Every week's duties, times 52, shared across everyone on the link — Sunday overtime included. "Days at work a year" is Monday to Saturday only: the days a week × 365 ÷ 7. Fifty-two weeks are exactly two turns of this link, so these are everyone's figures whichever week they start on; today's are averages over its 20 weeks.

### Following the trains

For each hour, the share of the day's floor staff is set against the share of the day's train service in the December 2026 timetable, each train weighted by its length; the differences are squared — so one big mismatch counts far more than several small ones — added up and multiplied by 10,000. 0 would be a perfect match; lower is closer. The staff share uses the minutes each person spends on the floor, finer than the whole-person counts on page 5, where anyone on for part of an hour counts. It measures the shape of the day, not a staffing level: an extra person in a quiet hour makes it worse although nobody is worse off. The ticket office is taken out first: it is staffed to its opening hours, not to the trains, which is why its rows on page 5 carry no match figure.

### The ticket office

In the proposal, two people on every ticket-office shift, not floor cover except at the quiet ends: Monday to Saturday one of each pair is on the floor until 08:00 and from 19:30. On a Sunday one early is on the floor until 09:00; the two lates until 15:00, then both in the office, one back on the floor from 18:00 and the other from 22:00, once the office has shut at 21:40 and cashed up; a late that runs to 22:30 spends its last half hour on the floor. On a Saturday the lates stay until 22:30 to cash up after it shuts at 22:10 (today's roster: 22:00). Today one person keeps the office on a Sunday afternoon and evening, until 22:00.

### Rest, runs and weekends

The shortest gap is the single tightest one in the fixed duties, Saturday into Sunday and line into line included. The most days in a row counts Sunday overtime too; it is given for the fixed duties, with "up to" where a cover week placed as badly as it can be, its Sunday worked, would make it longer, and the 13-day limit is tested at that worst case. A full weekend off is a Saturday off then a Sunday off. The shortest gap is one gap: a rota with many rests of just over 12 hours reads the same as one with none.

### Handover

The overlap when one shift takes over from another. The December rules ask for 15 minutes to each closer, and 20 in the ticket office (30 on a Sunday) — page 6.

### Familiar shift times

How many of the shift times somebody already works today — a rough guide to how much there is to learn, not to what staff will accept.

### Fatigue warnings

The Office of Rail and Road's good-practice list of roster patterns that tend to tire people, with four rail-industry checks. A warning is a question to discuss, not a breach, and this is not a fatigue risk assessment. Early starts at a 06:20 station and a weekly rotation come with every link, today's included, so they are recorded on page 7 and not counted. In the ORR's checks an early start is one from 05:00 up to 06:59 and a "block" of earlies is two or more in a row; the start-time rows count how many times a start moves by more than two hours (a count, not hours); the week-to-week move is the average change in a week's mean start time, Sundays included and cover weeks left out.

### What is confirmed

The December 2026 staffing levels and the Sunday cover were confirmed verbally on 29 Sep 2026; the 26-line link, 5 cover weeks and the ceiling of 219 contracted days a year were set on 1 Oct 2026.

The rota is supplied with this PDF, in the pack's import files, as Balanced-V9-MB-26-V9-import.txt, ready to paste into the Links page (Import), which re-runs the hard limits, the fatigue checks, the weekends, the hours and the hour-by-hour staffing on it — giving the runs at their worst case, which pages 2 and 7 give as "up to" — though not the December rules or the yearly figures, which are this sheet's. The fingerprint **e2b7d945** in every footer is worked out from the 182 days of the rota and nothing else, so a printout always matches the design it came from.